

THE RECORDER TRACKS THE SCHEDULE: DETECTED IDENTITY-LOCK PERIOD FOLLOWS BLOCK LENGTH UNDER AN ACTIVE-BLOCK RECORDER, WITH A TWO-LAYER MEASUREMENT CANON FOR DIFFUSION-LLM BOUNDARY ANALYSIS

Anonymous authors

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ABSTRACT

Position-decomposed diagnostics of diffusion large language models can confuse sampler geometry with model behavior because an active-block recorder exposes positions only through the current sampling window. In a recorder-scoped comparison, we hold the model, questions, canvas, decoding rule, and total function evaluations fixed while applying a composite intervention: block partition changes and the required per-block step allocation necessarily co-varies. The detected boundary period follows block length on this bed; separately registered non-nested probes on a distinct canvas-96 bed do not reveal a recorder-observable fixed-32 component. We contribute a reporting canon that separates what boundary-set composition makes reachable before data from what measured magnitudes support after data. The amended decision branch, composite intervention, and unmeasured cross-model, cross-schedule, and cross-recorder scope preclude a model-mechanism claim.

1 INTRODUCTION

Diffusion language models generate discrete sequences by iterative denoising (Austin et al., 2021; Zheng et al., 2024; Lou et al., 2024; Shi et al., 2024; Nie et al., 2025). As decoding schedules become variable or structured (Li et al., 2025; Sun et al., 2026; Wang et al., 2026), a position-resolved diagnostic can inherit periodicity from the schedule that exposes its positions. Without an intervention, such a readout cannot distinguish model-intrinsic state, sampler state, and what the recorder makes observable.

We examine identity-lock time on a blockwise-decoded LLaDA-style model (Nie et al., 2025; 2026). An apparent period of 32 motivates a matched composite intervention over the block partition and its required per-block step allocation: the model, 300 questions, canvas, total NFE, decoding rule, and temperature remain fixed. The active-block recorder, however, logs only masked positions in the currently active block. Its lock-time profile is therefore defined relative to the sampling window (Figure 1).

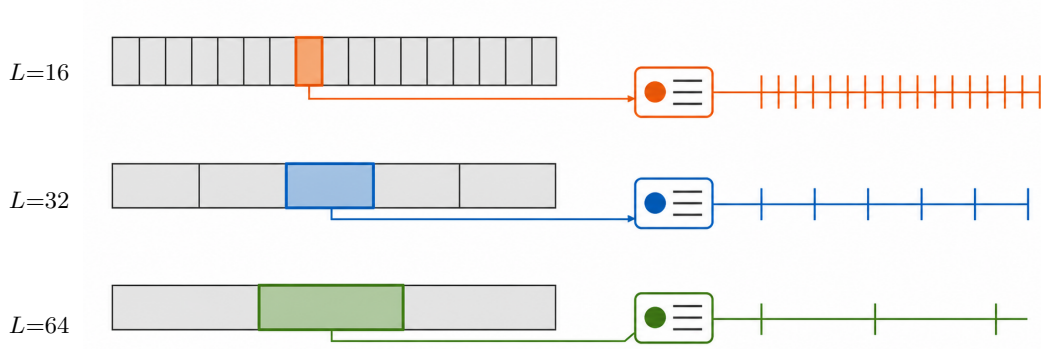


Figure 1: Qualitative measurement pipeline. From top to bottom, block length L grows while canvas extent stays fixed; color marks the one active block exposed to the recorder, and the right-hand ticks depict the boundary spacing visible through that window. Pulse and group counts are schematic, not quantitative evidence. Measured profiles and separations are in Figure 2 and Table 3.

Across the matched arms, the detected period follows the configured block length; separate non-power-of-two probes show the same recorder-observable pattern without the nested-32 boundary relation. Neither result identifies a causal subtype. The originally registered model-intrinsic branch was withdrawn after treatment (amendment v3; Appendix A), and the measured candidate-32 signs are not values forced by nesting alone.

Our contribution is one reporting rule: state before data which outcomes boundary-set composition makes reachable, then state separately what measured magnitudes show. It prevents a recorder-defined window from quietly becoming a claim about the mechanism that produced the tokens. A development control, matched block-length intervention, bootstrap stability checks, negative probes, and content-sensitivity analysis support this distinction on the tested beds; none establishes a mechanism across models or schedules.

2 MEASUREMENT AND DESIGN

Core notation is defined here before it is used; Appendix C retains the longer terminology crosswalk.

Notation. For decode i and relative canvas position r , $\ell_{i,r}$ is the recorded identity-lock time. Its positionwise median is $m(r)$ and $d(r) = m(r+1) - m(r)$. For candidate period P , $\mathcal{B}_P$ contains candidate boundary indices and $\mathcal{I}_P$ their interior complement; $\text{sep}(P)$ is the boundary-minimum minus the interior-maximum of $d(r)$, and P^* maximizes it. “Canvas 96” denotes the separate 96-position probe bed. $\text{BIN}k$ denotes a restricted content null with k equally populated offset-rank strata.

Model bed. We use an LLaDA-style masked dLLM at the LLaDA-8B-Instruct scale with a locally registered checkpoint on GSM8K (Cobbe et al., 2021). Decoding uses temperature 0 and `remasking=low_confidence`, the confidence-ordered parallel decoding rule introduced for conditional masked language models and used by LLaDA (Ghazvininejad et al., 2019; Nie et al., 2025). The main differential fixes NFE at 48 and uses the same 300 questions. Its L32/NFE48 arm is a frozen *development control*: the registered contiguous `decode_id` $\in [345, 645)$ subset of a pre-existing pool. The L32/NFE32 content bed is separate and supports only Section 3.4.

Setup and protocol. Table 1 consolidates every bed and schedule. The headline length metric is $d(r) = m(r+1) - m(r)$, the adjacent difference of median identity-lock time. For content, the same lock-time ordering is compared with residualized confidence under an explicit eight-stratum offset-rank null (BIN8; four-position granularity), plus a same-amplitude *position-fake*: confidence values are assigned to positions to preserve amplitude while destroying the observed content-to-position pairing. Intervals use cluster-bootstrap percentiles over units (Davison & Hinkley, 1997); decode resampling preserves multiplicity.

Element	Measured arms	L32 control	Non-nested probes	Content bed
Canvas / NFE	256/48	256/48	96/48	256/32
L ; per-block	$16 : [4] \times 8 + [2] \times 8$;	$32 : [8] \times 4 + [4] \times 4$	$24 : [12] \times 4$;	32
steps	$64 : [16, 16, 8, 8]$		$48 : [24] \times 2$	
Questions	same 300	contiguous [345, 645), $n=300$	same 300	GSM8K
Candidate grid	{8, 16, 32, 64}	{8, 16, 32, 64}	{8, 16, 24, 32, 48, 64}	—
Resampling	period $B=2000$; stability/sign $B=1000$	period $B=2000$	period $B=2000$	interval $B=500$; null $R=2000$

Table 1: Unified configuration. All beds use the same model family, task, temperature, and remasking rule. The control’s contiguous [345, 645) subset is not part of the non-nested probes. Those two arms form a separate canvas-96 setup with $n=300$, NFE 48, the displayed schedules and grid, and 2,000 cluster-bootstrap replicates. NFE denotes total function evaluations. At fixed total NFE, changing the partition also changes the required per-block step allocation; the design is therefore a composite intervention and does not isolate an allocation effect.

Identity-lock time. For decode i , relative canvas position r , and recorded step s , let $x_{i,r,s}$ be the predicted token and $x_{i,r}^f$ its final value. We define

$$\ell_{i,r} = \min\{s : x_{i,r,t} = x_{i,r}^f \text{ for every recorded } t \geq s\}. \quad (1)$$

Thus, $\ell_{i,r}$ is the earliest recorded step after which the token identity never changes again. The position profile is $m(r) = \text{median}_i \ell_{i,r}$ and its adjacent difference is $d(r) = m(r+1) - m(r)$. All analysis uses relative generated-canvas position, not prompt position.

Min-max period separation. For each candidate period in the registered grid (Table 3), define the candidate boundary set $\mathcal{B}_P = \{r : r \bmod P = P-1\}$ and interior $\mathcal{I}_P$ as its complement. The registered estimator is

$$\text{sep}(P) = \min_{r \in \mathcal{B}_P} d(r) - \max_{r \in \mathcal{I}_P} d(r), \quad P^* = \arg \max_P \text{sep}(P). \quad (2)$$

Exact ties select the first (smallest) period in the ordered registered grid. Positive separation requires every candidate boundary jump to exceed every interior jump. On a strictly L -supported profile it penalizes divisors with false boundaries and multiples that leave true boundaries inside; without that condition it is not a general divisor/multiple guarantee. A period is detected only under the conjunction $\text{sep}(P^*) > 0$ and a cluster-bootstrap modal period equal to P^* with mode fraction at least 0.95. The same conjunction classifies each bootstrap readout below. We resample the 300 decodes with replacement, preserving multiplicity, for $B=2,000$ replicates.

Selection stability and its honest reading of “no misses.” Two resampling readouts are reported. First, a *joint paired cluster bootstrap* resamples a shared index set of the 300 decodes once and applies the same indices to all five arms (the matched three plus the two canvas-96 probes), so the headline is tested as a *cross-arm relation* on shared questions rather than as five independent marginals. Across $B=2,000$ replicates (seed 20260824), $\Pr[P_{16}^*=16 \wedge P_{32}^*=32 \wedge P_{64}^*=64] = 1.0000$; each of all five arms also has marginal $\Pr[P^*=L] = 1.0$ (Table 11). Second, we report the probability that the detection rule actually fires, $\Pr[\text{sep}(L) > 0]$, instead of the “0 misses” language. Table 2 consolidates this readout across all five arms. Because the lock times are integer (temperature 0), the min-max separation is an integer-valued extremum of integer medians, so the bootstrap lower bound can sit exactly at the point estimate and “0 misses” reflects discrete selection stability rather than an absence of error. We therefore do not read a 0/2,000 miss count as “no false detections.”

Because sep is an extremum statistic, we also report a *non-extremal* robustified comparison: a rank-sum (Mann-Whitney U) separation between candidate-boundary and candidate-interior values of $d(r)$. Its argmax period equals L on *all five* arms, including the L32 control where the min-max separation was weakest; the per-arm statistics are in Table 2 and their provenance in Table 11. A quantile-based separation survives only on L64 and L48 and selects multiples otherwise, which we report honestly as partial rather than as a full vote for the extremum statistic.

Replication accounting. Table 1 is the complete resampling inventory. Period selection and its interval share one draw, whereas the sign and stability checks use separate draws; Table 3 identifies the shared readouts. Uncounted values are observed-data point estimates; artifacts are in Table 11.

Matched intervention and scope. The two treatment arms and frozen development control are in Table 1. At fixed total NFE, changing the partition necessarily changes per-block step allocation, so the treatment is partition-plus-required-allocation. A within-arm rank transform preserves $P^*=L$ and each candidate-32 sign on all five tested block-length arms, including the non-power-of-two $L=24$ and $L=48$ (Table 11); a uniform-allocation arm was not registered. The result therefore identifies a recorder-output association with the composite partition-plus-allocation schedule, not which component places or scales the boundary. A factorial allocation/partition intervention is future work.

Registration and two-layer reachability. The estimator and detection rule were fixed before treatment, but amendment v3 withdrew the model-intrinsic decision branch after treatment; the timestamped chronology is in Appendix A, Table 10. The canvas-96 probes were registered separately before their execution. We therefore report reachability at two layers. *A priori*, nesting such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ determines set composition but not the sign of $\text{sep}(32)$. *From events*, the observed magnitude ordering gives $\text{sep}(32) = -2$ on L16 and a negative value on L64. Thus a recorder-observable fixed-32 component is not detected under these active-block regimes; this does not identify model-intrinsic or sampler-state causes. The exact loci and synthetic condition are consolidated in Table 6; the legacy verdict labels remain only in Appendix C.

Separately conditioned content analysis. On the separate L32/NFE32 bed, BIN8 permutes confidence within eight equally-populated offset-rank bins after a fixed nuisance fit. The initial confidence `conf0` is read at the first recorded step of a position’s block, following confidence-ordered parallel decoding (Ghazvininejad et al., 2019; Nie et al., 2025). This supporting diagnostic is not a cross-block-length validation.

3 RESULTS

3.1 THE DETECTED PERIOD FOLLOWS BLOCK LENGTH

The development control recovers its known period, and both treatment arms move to their respective block lengths (Figure 2; Table 3). Only the selected period has positive min–max separation, and the bootstrap selects it in every replicate. The paired cross-arm relation holds in all 2,000 shared-cluster resamples, while a rank-sum point estimator independently selects $P^*=L$ in all five arms. Across the five arms, P^* leads the best wrong candidate by +2.0 to +8.0 d -units; four reach 5% significance against the within-block null (exact $p_{\text{margin}} \leq 2.1 \times 10^{-2}$), while L24 is marginal at $p_{\text{margin}} = 7.1 \times 10^{-2}$ (Table 5). The alignment is therefore not reproduced by within-block jitter under that registered grid.

Readout	L32		L24		L48
	L16	control	L64	probe	probe
$\Pr[\text{sep}(L) > 0]$, joint bootstrap	1.0	0.8805	0.9995	0.999	1.0
Rank-sum z at L	13.9	8.3	4.4	3.9	2.0
Wrong-candidate fire rate, original three-bed probe	0.0	0.0	0.0	—	—

Table 2: Selection stability and non-extremal robustness (not a false-positive rate). The first row uses one $B=2,000$ joint paired bootstrap and reports how often the conjunction $\text{sep}(L) > 0$ and mode fraction ≥ 0.95 fires. The L32 control’s 0.8805 is the weakest arm and appears alongside the perfect modal-argmax relation. The second row is the Mann–Whitney comparison, whose maximizing candidate is L on every arm. The final row is a separate $B=1,000$ in-place wrong-candidate diagnostic for the original three beds: its zeros follow from negative point estimates and are not fresh-null false-positive rates. Within an arm, cells share draws. Provenance is in Table 11.

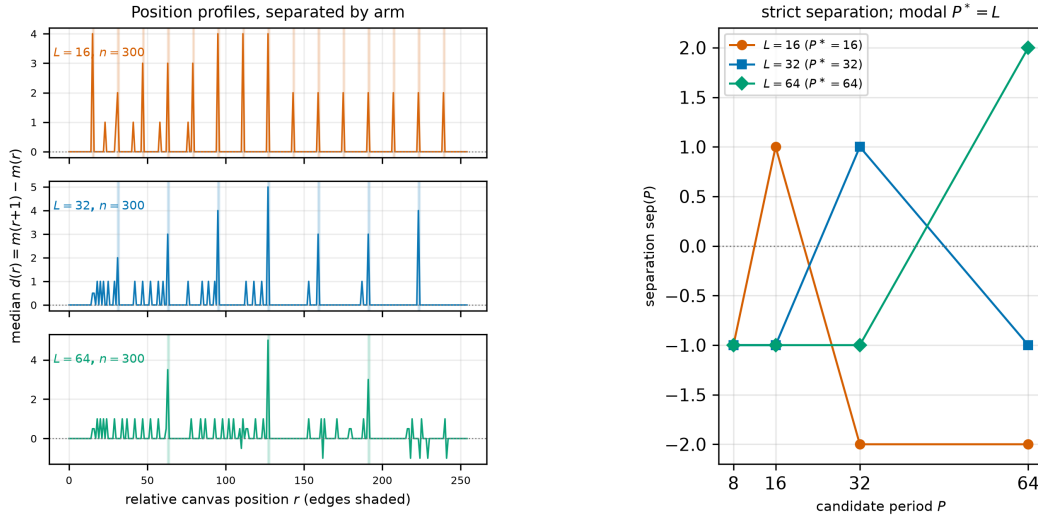


Figure 2: Data-derived differential result. Left: three stacked facets isolate the median adjacent-difference profile $d(r) = m(r+1) - m(r)$ for L16 (orange), the L32 frozen control (blue), and L64 (green), each from $n=300$ decodes; shaded ticks mark that arm’s block edges. The control is the contiguous `decode_id` $\in [345, 645)$ subset. Right: only the matching candidate has positive separation in each arm. Regenerated by `scripts/r265_fig_dprofile_overlay.py`; quantitative values come from the registered events, and the recorder-level comparison does not distinguish sampler from model mechanism.

L (source)	$\text{sep}(P)$				P^* (mode frac.)	$\text{sep}(P^*)$ stability band [†]	$(\min_B, \max_I)$		
	$P = 8$	$P = 16$	$P = 32$	$P = 64$			@ P^*	$ B_{P^*} $	$ I_{P^*} $
16 (measured)	-1	+1	-2	-2	16 (1.0)	[+1, +1]	(+2.0, +1.0)	15	240
32 (control)	-1	-1	+1	-1	32 (1.0)	[0, +2]	(+2.0, +1.0)	7	248
64 (measured)	-1	-1	-1	+2	64 (1.0)	[+1, +3]	(+3.0, +1.0)	3	252
24 (probe)	—	—	—	—	24 (1.0)	[+1.5, +4.0]	(+5.0, +1.0)	3	92
48 (probe)	—	—	—	—	48 (1.0)	[+4.0, +6.5]	(+8.0, +1.5)	1	94

Table 3: Development-control, differential, and probe results. Each arm uses 2,000 cluster-bootstrap replicates; the mode fraction and percentile band are paired readouts of the same draw. The $(\min_B, \max_I)$ columns report the boundary minimum and interior maximum, followed by their observed set cardinalities. Every arm selects its configured block length. Because lock times are integer-valued, the bands diagnose selection stability rather than coverage: L16 collapses to a point mass, while the L48 probe rests on one boundary location. Tables 1, 2, 5, 7, and 8 give the complete resampling inventory and separately budgeted checks; provenance is in Table 11.

Candidate 32 under the active-block recorder. Table 3 reports $\text{sep}(32) = -2$ on the measured $L = 16$ arm. The exact values in Table 6 show why: candidate-32 boundary values remain positive, but the measured odd-16 interior is larger. The joint sign $\text{sep}(16) > 0 \wedge \text{sep}(32) < 0$ holds in all 1,000 dedicated bootstrap replicates. Candidate 32 is therefore not detected at the recorder-observable layer on this bed. Nesting alone does not force that sign, and the result says nothing about an unobserved model-intrinsic or sampler-state component.

Probe arm	Non-nested sep(32)	Disclosed marginal candidate
$L=24$ probe	-8.0	$P=48$: -1.0; band $[-3.0, 0.0]$
$L=48$ probe	-7.5	—

Table 4: Separate canvas-96 non-nested probes. Candidate 32 is disjoint from the $L=24$ boundaries. The L24 marginal cell concerns the mismatched $P=48$ candidate rather than detected $P^*=24$; the displayed percentile band is a selection-stability diagnostic, not a coverage-valid interval.

Arm	P^*	sep(P^*)	m_{obs}	p_{sepL}	p_{margin}
L16 (canvas 256)	16	+1.0	+2.0	5.0×10^{-4}	1.5×10^{-3}
L64 (canvas 256)	64	+2.0	+3.0	5.0×10^{-4}	1.0×10^{-3}
L24 (canvas 96)	24	+4.0	+5.0	5.0×10^{-4}	7.1×10^{-2}
L48 (canvas 96)	48	+6.5	+8.0	2.3×10^{-2}	2.1×10^{-2}
L32 control (canvas 256, NFE 48)	32	+1.0	+2.0	5.0×10^{-4}	1.0×10^{-3}

Table 5: Separation-margin selectivity calibration. Each of $R=2,000$ replicates (seed 20260825) permutes $d(r)$ within each true L -block, including its edge index, then records the argmax candidate’s top-minus-second gap. Here $m_{\text{obs}} = \text{sep}(P^*) - \max_{c \neq P^*} \text{sep}(c)$ and the discrete exact $p = (1 + \#\{m_{\text{perm}} \geq m_{\text{obs}}\})/(R + 1)$. The p_{sepL} column tests $\text{sep}(P^*)$ alone; p_{margin} is the selection-controlled readout. Provenance and the original-gate size audit are in Table 11 and Appendix A.4.

Measured class	Loci	$d(r)$ values	Readout
L32 true-32 boundaries	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 5, 3, 3, 4]	development control
L16 true-16 odd chain	[15, 47, 79, 111, 143, 175, 207, 239]	[4, 3, 3, 4, 2, 2, 2, 2]	candidate-32 interior max 4
L16 would-be $\mathcal{B}_{32}$	[31, 63, 95, 127, 159, 191, 223]	[2, 3, 4, 4, 2, 2, 2]	boundary min 2; sep(32) = -2
L64 mid-block 32	[31, 95, 159, 223]	L64: [0, 0, 0, 0]; L32: [2, 4, 3, 4]	recorder-observable collapse
L64 true-64	[63, 127, 191]	L64: [3.5, 5, 3]; L32: [3, 5, 3]	true boundary remains
Strict-support synthetic L Source periods Selected P^*			
	16	{8, 16, 32, 48}	16 for every source
	32	{8, 16, 32}	32 for every source
	64	{8, 16, 32}	64 for every source

Table 6: Consolidated negative evidence, reachability, and development check. A priori, nested boundary sets determine membership but do not force the sign. From events, L16 candidate 32 has boundary minimum 2 and interior maximum 4; the joint sign holds in 1,000/1,000 bootstrap replicates. On L64, would-be 32 boundaries are recorder-observable interiors. The lower panel is a development-only synthetic control: $P^* = L$ holds for the shown strictly L -supported profiles, not for arbitrary residuals. The upper rows are observed-data readouts from their arm samples, not additional replications. Provenance is in Table 11.

3.2 SEPARATION AND STABILITY, NOT AN EFFECT-SIZE POWER CLAIM

The central estimand is discrete period identity. On the observed samples, positive min-max separation is a deterministic property of the recorded profiles; the bootstrap then probes the stability of period selection to resampling decodes. The selected period is stable under both the per-arm bootstrap and the joint paired bootstrap that re-applies one shared question-index draw to all five arms

(Section 2); the detection rule fires in 0.88–1.0 of resamples across arms, with the L32 development control the weakest at $\Pr[\text{sep}(32)>0]=0.8805$. A robustified non-extremal rank-sum separation still selects L on all five arms. This does not establish an arbitrarily small effect or a universal period: it establishes stable selection among the registered candidates on the stated bed, and the low-factor cells ($|B_{48}|=1$ on the L48 probe) temper any margin reading. The L32 control also reuses a pre-existing contiguous [345, 645] pool and its percentile stability band $[0, +2]$ touches zero (it is not a coverage-valid CI). These three weaknesses are disclosed together; a fresh in-session L32 rerun is GPU-gated future work.

3.3 A TESTED SPECTRAL RESIDUAL FAMILY DOES NOT TRIGGER ON L16

For a nested candidate such as $\mathcal{B}_{32} \subset \mathcal{B}_{16}$, a *negative* $\text{sep}(32)$ is not evidence of absence: the extremum comparison cannot recover every possible residual shape. An *exploratory* (post-data, not pre-registered) check therefore asks whether a *non-nesting* estimator recovers a residual source period that sep misses. We use a DFT periodogram (non-nesting): for each candidate P , $G(P) = \|\sum_r d(r)e^{-2\pi ir/P}\|/\|d\|$ is the normalized spectral magnitude of the adjacent-difference profile at frequency $1/P$, with a permutation null (shuffle $d(r)$ positions; 4,000 replicates) and family-wise q_{95} over all candidates. Harmonic testing follows the periodogram tradition (Fisher, 1929); our finite-grid threshold is a bespoke single-step maximum over candidates, using the standard permutation max-statistic construction (Nichols & Holmes, 2002). We also run a direct two-sample test of the two $L=16$ -boundary subsets in Table 6 (the odd-16 chain vs. the would-be- $\mathcal{B}_{32}$ positions), a reachable test of residual 32-amplitude modulation.

Case	$G(8)$	$G(16)$	$G(24)$	$G(32)$	$G(48)$	$G(64)$	$q_{95}@32$
Synthetic, residual-32							
present	3.58	3.58	—	4.86	—	—	1.74
Synthetic, no residual	4.01	4.01	—	0.25	—	—	1.75
Real L16 measured	3.79	3.70	0.15	0.23	0.36	0.18	2.15 (FW)

Supporting check	Statistic	Threshold / reference	Reading
Injected residual	Residual-32 amplitude 0.5 over comb-16 amplitude 3.0; $G(32)=4.86$	$q_{95}=1.74$	fires
No-residual synthetic	$G(32)=0.25$	$q_{95}=1.75$	null
Real L16 remaining	$G(24)=0.15$, $G(48)=0.36$, $G(64)=0.18$	per-candidate $q_{95} \approx 1.71\text{--}1.78$	no residual candidate
Comb harmonic	$G(8)/G(16)=1.000$ (pure comb), 1.024 (real L16)	1.0 reference ratio	recorder-comb harmonic
Power sweep	Amplitude 0.15: no fire; 0.20: fire; grid step 0.05 (5.0%–6.7% of comb amplitude 3.0)	q_{95} above	transition bracketed
Odd-16 vs. would-be- $\mathcal{B}_{32}$	Mean $ \Delta d =0.036$; permutation $p=1.0$	$\alpha = 0.05$	no amplitude split

Table 7: Spectral (DFT) periodogram of the adjacent-difference profile $d(r)$ — an *exploratory*, non-nesting check (Section 3.3). $G(P)$ is the normalized spectral magnitude at frequency $1/P$; “fires” means the 32-component exceeds its permutation q_{95} (4,000 replicates). The upper panel completes the synthetic diagnostic grid and reports every real candidate; the lower panel separates the supporting power, harmonic, and two-sample checks from their interpretation. The synthetic residual fires, whereas the comb-only synthetic and real L16 profile are null at the target period. On real L16, no candidate in the tested grid is significant; only the recorder-comb harmonics fire. The power sweep brackets sensitivity for the tested sinusoidal family only, and the direct boundary-subset comparison finds no split. Provenance is in Table 11.

Development control. On the recorder-consistent synthetic grid, with a clean sinusoidal residual-32 component overlaid on the $L=16$ comb, sep returns $P^*=16$ (blind) while the spectral 32-component fires; on the no-residual synthetic comb, both are silent (Table 7). This is a positive control for the tested sinusoidal family, not validation against arbitrary residual shapes.

Real L16 arm. The tested candidate grid does not exceed its family-wise threshold at $P = 32$, and the boundary-subset test finds no amplitude split. The registered power sweep brackets the tested-family transition and records its amplitude range in Table 7. Hence the measured arm does not trigger for this tested sinusoidal family and amplitude range; arbitrary residual shapes are not ruled out (Table 11).

Canvas-96 non-nested probe (boundary disclosure). On the separate canvas-96 probe beds, candidate 32 is *non-nested* ($\mathcal{B}_{32} \cap \mathcal{B}_{24} = \emptyset$), and the same estimator does not detect a recorder-observable fixed-32 component without relying on the canvas-256 nesting relation. Table 3 reports the two probe estimates, their selected block lengths, and the disclosed marginal mismatched cell (Table 4); the probes used about 0.38 GPU-h. This is a scoped extension (canvas 96 $\neq$ 256), not a replacement for the matched canvas-256 arms.

The L64 mid-block reversal and its L32 comparator are reported without a separate verdict vocabulary in the consolidated negative-evidence Table 6.

3.4 A CONTENT ASSOCIATION ON A SEPARATE SHARED BED

Estimand τ . On the separate L32/NFE32 bed, one unit is a decode-block with resolved positions $\mathcal{R}$; c_r is initial confidence and $o_r = r \bmod 32$ is block offset. The estimand is the median per-unit rank association after residualizing confidence on offset:

$$r_u = \frac{\sum_{p < q \in \mathcal{R}: \ell_p \neq \ell_q} \text{sgn}(\ell_q - \ell_p) \text{sgn}(\tilde{c}_p - \tilde{c}_q)}{\#\{p < q \in \mathcal{R} : \ell_p \neq \ell_q\}}, \quad \tilde{c}_r = \text{rank}(\text{rank}(c_r) - \hat{\beta} \text{rank}(o_r)), \quad (3)$$

Here the registered statistic fits $\hat{\beta}$ once on the observed data and shuffles the resulting $\tilde{c}$ within offset-rank strata. A sensitivity check instead refits $\hat{\beta}$ inside every permutation replicate. Earlier lock with higher residualized confidence contributes +1. Inference is rooted at decode clusters: the 500-replicate interval resamples decodes and carries their block units with multiplicity. The position-fake keeps the observed amplitude but reassigns it to positions, destroying the observed content-position pairing. This differently budgeted bed supplies supporting, not cross-block-length, evidence.

Quantity	Estimate / band	Null q_{95}	Null readout
Content-layer τ_{res}	0.2722 [0.2649, 0.2812]	BIN16 0.20235	0/2000 exceed
Refit- $\hat{\beta}$ sensitivity	0.2722	BIN8 0.13545	0/2000 exceed
Same-amplitude position fake	max. margin +0.00979	observed margin +0.12837	below observed

Table 8: Content-versus-structure result on the separate $L = 32$, NFE-32 bed. The statistic τ is the median decode-block rank association between earlier lock and higher confidence after offset residualization. BIN16 is the load-bearing conservative, non-degenerate null; the per-replicate nuisance refit is a separate BIN8 sensitivity and replays bit-exactly. Table 13 reports the full granularity ladder and three-bucket one-flip classification. This supporting analysis is not a cross-block-length experiment.

The full-granularity ladder in Table 13 is essential. The observed association clears every non-degenerate restricted null through BIN16, but matches the zero-power point mass at BIN32. Thus content association is established at moderate granularity, never at the finest registered granularity; the construction-specific BIN32 line is not interpreted as a lower bound. Its behavior under $L \in \{16, 64\}$ is unmeasured, and the two analyses are not claimed to be mutually calibrated.

The full operational checklist and its strictly- L -supported scope are given in Appendix A.

4 LIMITATIONS

- **Observation is not mechanism.** The active-block recorder cannot separate model-intrinsic state, sampler state, and recorder-observable state. Candidate 32 is not detected only at the last layer under the tested regimes.

- **Registration history.** The estimator was selected with a failed development control before treatment, but the model-intrinsic decision branch was withdrawn after treatment. The non-nested probes have their own pre-run registration and a different 96-position canvas.
- **Bed and grid.** Evidence covers one model family, one task, and the candidate grids in Table 1; off-grid periods, other models, tasks, samplers, and untested block lengths are not covered.
- **Composite intervention.** Partition and required per-block step allocation vary together. Boundary location is measured; a distinct allocation effect on amplitude is not.
- **Boundary cardinality.** The min-max separation is an extremum statistic, so its support matters: on the probe arms the selected period rests on few measured boundary positions (the L48 probe has $|B_{48}|=1$), making its positive separation a single-locus readout rather than a stable margin. We report $|B_{P^*}|$ and $|I_{P^*}|$ per arm in Table 3 for this reason.
- **Recorder and control arms.** A dual-recorder comparison that logs the same trajectory through active-block and full-canvas/passive instruments is GPU-gated future work; no cross-recorder result is claimed. A fresh in-session L32/NFE48 run is also GPU-gated: the present control reuses a pre-existing contiguous pool, has the lowest positivity (0.8805), and its percentile stability band touches zero.
- **Content scope.** The content association comes from one L32/NFE32 bed and survives only through moderate null granularity (it is not established at the finest registered BIN32 granularity). Its behavior on the L16/L64 beds is not measured.

5 RELATED WORK

Foundational discrete-diffusion work studies the generative formulation or likelihood rather than this recorder trace. Austin et al. (2021) define structured discrete-state denoising; Zheng et al. (2024) change its parameterization; Lou et al. (2024) estimate data-distribution ratios; and Shi et al. (2024) simplify masked diffusion. In each comparison, the object and observation window are the model/sampler formulation and its generated trajectory, not an active-block event recorder perturbed while model and NFE stay fixed. We therefore use them as background, not as failed instances of our test.

Work	Relevant decoding intervention	(a) position-resolved commit/lock timing?	(b) controlled block length?
Gu et al. (2018)	Fully parallel NAT with fertility planning	No	No
Wang et al. (2018)	Group size K controls semi-AR parallelism; reports positionwise cross-entropy, not commit timing	No	Yes
Ghazvininejad et al. (2019)	Iteratively remasks least-confident tokens	No	No
Nie et al. (2025)	LLaDA low-confidence remasking	No	No
Nie et al. (2026) (preprint)	Improved LLaDA inference/training	No	No
Li et al. (2025) (preprint)	Varies denoising length, not block length	No	No
Sun et al. (2026) (preprint)	Adapts decoding structure by Bayesian inference	No	No
Wang et al. (2026) (preprint)	Revisits earlier diffusion decisions	No	No
This work	Fixed-NFE composite block-partition plus required-allocation intervention	Yes	Yes, composite

Table 9: Verified blockwise and parallel-decoding comparison. Column (a) requires position-resolved *commit* or *lock timing*, not merely positionwise loss; column (b) requires an explicit block-length intervention, not a change in total denoising length or an adaptive schedule. “Yes, composite” discloses that our per-block allocation necessarily co-varies with the partition.

The nearest methods on the schedule axis make the distinction sharper. Fully parallel NAT (Gu et al., 2018), semi-autoregressive grouped decoding (Wang et al., 2018), confidence-ordered mask-predict decoding (Ghazvininejad et al., 2019), LLaDA-style decoding (Nie et al., 2025; 2026), and variable or reversible schedules (Li et al., 2025; Sun et al., 2026; Wang et al., 2026) all change how positions are generated or revisited. Table 9 asks the narrower question: does each work report position-resolved commit/lock timing, and does it vary block length as a controlled intervention?

Among these verified works, none reports position-resolved commit/lock timing across a controlled block-length intervention. The recorder-scoped differential in Section 3.1 is therefore new on that axis, while Wang et al.’s positionwise cross-entropy periodicity is the closest partial precedent. These setup differences do not imply a field-wide absence claim. Appendix Table 14 gives the fuller row-wise comparison. Section 3.4 also differs from confidence-dynamics work: it measures association between the first-step confidence and later recorder lock ordering, rather than proposing a new confidence-based remasking policy.

A second, closer axis is the measurement literature that already warns that a readout instrument can induce the signal it claims to observe. Sahoo et al. (2026) show that a linear probe’s near-perfect accuracy on language-model hidden states can be driven entirely by task-format confounds (source identity, option count, response length), vanishing after deconfounding — an instrumentation-bias lesson we mirror in the recorder-window geometry. Jiang et al. (2020) argue that any prompt-based measurement is at best a *lower bound* on stored knowledge because a sub-optimal prompt fails to retrieve facts the model does hold; this is exactly the a-priori-reachability vs. measured-magnitude split we make explicit. Belinkov & Glass (2019) survey how the choice of analysis method changes what is measured about a representation, and Feng et al. (2024) show a model can internally encode a faithful world state while decoding unfaithfully — the observational-equivalence gap between a recorded layer and the behavior that produced it. Conversely, Stolfo et al. (2023) attribute mechanism from internal states via causal mediation; we deliberately do *not* — our contribution is the reporting rule, not a mechanism claim. To make the boundary checkable, the appendix setup comparisons (Appendix A) state each prior object’s observation window, intervention, and measurement-layer contrast. Table 15 makes the same-dimensionality question explicit: each closest measurement work operates on a scale other than recorder-window period separation, so our addition is a differential separator and negative probes on that dimension, not a claim to out-measure those works on their own scale.

More generally, construct validity requires matching an instrument to the quantity its coefficient or probe can support: conditioned coefficients can answer a different target than the headline exposure (Westreich & Greenland, 2013), and representation analyses inherit the choices and confounds of their probes (Belinkov & Glass, 2019; Sahoo et al., 2026). Preregistration separates planned analysis from outcome-contingent interpretation but does not itself validate a construct (Nosek et al., 2018). Our two-layer rule adds a precise audit record rather than a general validity theory: before events, freeze the set-algebraic outcomes the estimator makes reachable; after events, report the magnitude ordering without retroactively treating reachability as measured evidence. The post-treatment amendment remains visible, so the split does not convert it into an a-priori claim.

This work is therefore not a decoder or a mechanism attribution. Its contribution is the two-layer reporting canon, exercised with a matched recorder-level differential. The min-max estimator separates the displayed divisors and multiples only under the stated strictly- L -supported condition; the exploratory DFT control covers only the tested sinusoidal residual family.

6 CONCLUSION

Under the tested active-block recorder, the detected identity-lock period follows block length. Development controls, matched arms, non-nested probes, bootstrap stability, and negative evidence support that recorder-level statement. They do not distinguish model-intrinsic state, sampler state, and recorder observability. The transferable result is the reporting rule: state what is reachable from set composition before data, then separately state what the measured magnitudes show. Calibrate position-decomposed measurements to the schedule that recorded them, and do not infer a mechanism or cross-bed relation that the intervention did not measure.

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A PROTOCOL CHRONOLOGY, MEASUREMENT CANON, AND SENSITIVITIES

A.1 REGISTRATION CHRONOLOGY

Timestamp	Status	Recorded event
2026-08-20 17:55	Pre-data	Original registration filed before any GPU submission.
2026-08-20 18:09	Pre-data	Amendment v2 replaced the failed signed mean-difference score with the min–max estimator and registered the detection rule.
2026-08-20 23:07	Treatment	Both treatment arms ran (0.73 GPU-h).
2026-08-21 17:36	Post-data	Amendment v3 withdrew the model-intrinsic branch.
2026-08-21	Pre-data (probe)	Non-nested canvas-96 probe registration and amendments preceded its own run (0.38 GPU-h).

Table 10: Registration and amendment chronology. The estimator and detection rule were fixed before treatment; the model-intrinsic decision branch was withdrawn after treatment data were observed, so the chronology documents a scope change rather than a pre-treatment mechanism verdict.

A.2 ARTIFACT INDEX

Table 11: Claim-to-artifact map. This is the sole table that prints workspace-relative artifact paths; all narrative references point here. Full provenance continues in `evidence/claim_evidence_map.tsv`, `evidence/method_provenance.tsv`, and `evidence/experiment_registry.json`.

Claim / role	Workspace-relative artifact	SHA-16
L32 development control	results/blocklen_probe/G0_VERDICT_r3.json	984b4ff92fe462ad
L16/L64 period	results/r29_l16l64_sawtooth.json	d462dff346604321
Cross-arm bootstrap, rank-sum, stability bands	results/r295_joint_bootstrap_extremum_robust.json	9a40c56c7c7a8243
Within-block selectivity gate and size audit	results/decomposition_control_r314/r316_sep_withinblock_permnull.json; results/decomposition_control_r314/r320_margin_selectivity.json	f623432c139e440c; e7d1c3115253589e
Boundary/interior cardinalities	results/decomposition_control_r314/r316_bcardinality_per_arm.json	e3a1c7b2b3c6915a
Content statistic and interval	results/r10_poslevel_gates/r10_poslevel_gates_v2.json	0af4f3c03e6c6f6d
Content null ladder	results/r274_bin1632_null_ext.json	b5d91fdc0b827018
Per-permutation nuisance refit	results/r259_beta_recompute/r259_beta_recompute.json	d3b27b3c87c924ce
Bit-exact nuisance-refit replay	results/r259_replay_B/r259_replay_B.json	85b875d858d0db19
One-flip three-bucket classification	results/r311_oneflip/r311_oneflip.json	47735637e3777a3d
Position-fake control	results/r14_margin_sweep/r14_margin_sweep.json	91da8f8b91a6b6b9
L16 measured nested-candidate check	results/r58_z1b_L16_truepair.json	10530a1b478ff492
L64 mid-block comparison	results/r58_z2b_l64_neg_evidence.json	470908f1400e7a1c
Reachability table	results/r147_reachability_table/r147_reachability_table.json	52a983b1c2fade9e
Strict-support synthetic diagnostic	results/r147_entailment_demo/r147_entailment_demo.json	ee4f1ad85d5492a2
Spectral family and power grid	results/r259_non_nesting_spectral/r259_non_nesting_spectral.json; results/r267_spectral_power_table.json	a1b52a418ace4e07; 33768616bfa41ecc
Rank-transform sensitivity	results/r265_rank_transform_invariance.json; results/r299_orthogonal_separation_r299.json	f1760ce13789a67d; d90a4d56635f3771
Figure 2 manifest	results/r265_fig_dprofile_manifest.json	40079e3c19996a01
Control subset and row checks	results/r269_l32_control_subset_checks.json	e5455d953fad8367
Cross-environment reproduction	results/r269_estimator_byte_and_arm_recheck.json	2c5f824c0ef215db
Image-generation audit	build/image_generation_record.json	c019dbf2309f460b

A.3 MEASUREMENT CANON

For blockwise decoding observed through an active-block recorder, a position-resolved analysis should:

1. record block length, per-block steps, and generated-canvas coordinates;
2. calibrate within-block windows to the generating run rather than a fixed model constant;
3. report reachability twice: *a priori*, what estimator algebra and boundary-set composition imply; *from events*, what the measured magnitude ordering implies. In particular, $\mathcal{B}_{32} \subset \mathcal{B}_{16}$ does not force $\text{sep}(32) \leq 0$;
4. perturb the schedule when attributing a position-resolved diagnostic;
5. test content claims against a structure-preserving null and report its granularity; and
6. mark cross-bed relationships unmeasured until the corresponding intervention is run.

This is a measurement canon, not a mechanism claim. The registered estimator returns $P^* = L$ on the displayed *strictly L-supported* profiles: non-degenerate jumps occur on all L boundaries and no L -multiple's exclusive-boundary residual outscore the L -boundary minimum. It is not a guarantee for arbitrary residual profiles, and identifying commit internals would require instrumentation of sampler-state transfer.

A.4 SELECTIVITY DISCLOSURE (PERMUTATION-GATE SIZE AUDIT)

Original-gate size. The original within-block threshold was anti-conservative. Across the five arms, $10/25 = 40\%$ of wrong candidates (non-true- L grid points) were flagged as detected, so the nominal family-wise error of at most 5% was not realized. Table 12 gives the per-arm audit. The cause is that edge-including within-block permutation destroys the non-target global level/trend structure of $d(r)$, shifting the null downward and loosening the threshold.

Corrected margin and cross-condition red-team. Table 5 replaces that level gate with the top-minus-second separation margin and a discrete exact p . As a known-clean red-team, globally (cross-block) shuffling the same $d(r)$ constructs a structureless reference. The corresponding probabilities are consolidated in Table 12: four of five cross-condition checks are below 5% and L24 is again marginal. This supports genuine schedule-aligned structure rather than within-profile jitter. A genuinely clean arm has no blocks to permute, however, so this red-team is a cross-condition comparison, not a per-arm data-shuffle false-positive rate.

Arm	Original-gate wrong candidates	Cross-condition p_{margin}
L16	4/5	5.0×10^{-4}
L64	3/5	2.0×10^{-3}
L24	0/5	5.5×10^{-2}
L48	0/5	1.8×10^{-2}
L32 control	3/5	1.5×10^{-3}
Aggregate	10/25 (40%)	4/5 below 5%

Table 12: Selectivity-gate audit by arm. The original threshold flags 10/25 wrong candidates, whereas the cross-condition margin red-team is below 5% in four of five arms; L24 is marginal. These cross-condition probabilities are not per-arm data-shuffle false-positive rates.

Reporting rule. Any threshold-type permutation gate used here for selectivity reports three quantities together: the wrong-candidate hit rate, a known-clean-arm or explicit cross-condition check, and the discrete exact p . The 10/25 audit and the margin correction therefore accompany every selectivity claim based on this within-block gate.

A.5 THREE-PART EVIDENCE DECOMPOSITION

For auditability, we separate the recorder-relative object $d(r)$, the pre-treatment min-max decision procedure, and the scoped empirical finding that P^* follows L on the tested recorder beds. The shared-cluster bootstrap and rank-sum check support the decision procedure; the non-nested canvas-96 probes bound the empirical finding. Table 11 maps each component to its verifier. This decomposition is a falsification aid, not three independent novelty claims or a mechanism attribution.

A.6 CONTENT ACCOUNTING AND NULL GRANULARITY

Accounting / sensitivity check	Recorded value	Interpretation
Content-resolved block-units	6,178 of 6,274 candidates	Terminal-resolved units excluded
Resolved comparable pairs	1,571,854	Both positions resolve
Within-unit support	Median 4 distinct locks; median resolved-pair fraction 0.575	Unit-level order is observed
Resampling	$n_{boot}=500$; $R_{null}=2000$	Decode-cluster interval and restricted null
K-ladder restricted-null q_{95}	$K=1$: 0.00486; BIN4 0.09276; BIN8 0.14384; BIN16 0.20235; BIN32 0.27221 (point mass)	With $\tau_{res}=0.2722$, margins at $K=1$, BIN4, BIN8, and BIN16 are 0.26735; 0.17945; 0.12837; 0.06986; BIN32 is zero-power by construction and carries no lower-bound reading
Nuisance- $\hat{\beta}$ sensitivity	Refit inside each BIN8 replicate: $q_{95}=0.13545$; fixed-fit BIN8 $q_{95}=0.14384$	0/2000 exceedances; bit-exact replay. BIN16 remains the load-bearing conservative non-degenerate arm
One-flip sensitivity ($1/n_{selected}$)	$n_{selected}=6178$; all 6178 residual-sign flips give $\Delta=0$	DEGENERATE, by construction; no pass claim. The null-specification ladder, not this identity arm, carries the sensitivity conclusion

Table 13: Unit accounting and restricted-null sensitivity for Table 8. A block-unit is rooted in its decode cluster; decodes are resampled with multiplicity. The registered line fits the nuisance slope once; the displayed sensitivity refits it inside each replicate. One-flip granularity is $1/n_{selected}$, not permutation spacing $1/R$. Under the three-bucket contract, maximum single-point effect $\phi = 0$ is DEGENERATE (by construction) and carries no pass/fail claim; $0 < \phi \leq$ margin would pass and $\phi >$ margin would fail. BIN32 is likewise zero-power by construction. The full sweep therefore limits the positive finding to moderate granularity. Provenance is in Table 11.

A.7 CLOSEST-WORK SETUP COMPARISON

Table 14: Row-wise setup comparison for the cited nearest work. Entries state the object and observation window, the schedule/measurement action, and the resulting contrast with this recorder audit, including whether each reports *both* layers of our two-layer rule (a-priori reachability *and* from-events magnitude). These scoped contrasts do not assert that an entire research area lacks schedule diagnostics or mechanism analysis.

Work	Object and observation window	Schedule action / measurement	Measurement-layer contrast used here
Austin et al. (2021)	Structured discrete-state denoising; model/sampler trajectory	Schedule belongs to the formulation	Background; reports neither a-priori reachability nor from-events magnitude
Zheng et al. (2024)	Reparameterized text diffusion; generated trajectory	Reparameterizes a fixed generative process	Changes model parameterization; we hold the model fixed and perturb the recorder-facing partition
Lou et al. (2024)	Distribution-ratio estimation for discrete diffusion; likelihood/generation	Schedule supports the generative objective	Targets modeling/likelihood, not boundary-set reachability

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Table 14 continued

Work	Object and observation window	Schedule action / measurement	Measurement-layer contrast used here
Shi et al. (2024)	Simplified masked diffusion; model/sampler output	Simplifies the diffusion formulation	Supplies the masked-diffusion setup, not the two-layer recorder audit
Li et al. (2025)	Variable-length inference; whole-decode outcomes across lengths	Changes denoising length	Schedule change used for decoding; ours fixes total NFE over a block-partition co-movement
Sun et al. (2026)	Bayesian dynamically structured decoding; evolving decode	Adapts decoding structure	Schedule is the method’s action; ours treats it as a diagnostic perturbation
Wang et al. (2026)	Reversible diffusion decoding; revisable trajectory	Permits decisions to be revisited	Reversibility changes the trajectory; ours audits masked active-block events
Sahoo et al. (2026)	Hidden-state linear probe; language-model input/activations	Deconfounds measurement from format	Warns instrumentation confounds can induce a readout; reports measured magnitudes, not a-priori set composition
Jiang et al. (2020)	Prompt query of LM knowledge; output distribution	Optimizes prompts	Frames any measurement as a lower bound; distinguishes reachability from measured magnitude for facts
Belinkov & Glass (2019)	Analysis/probing methods for representations; activations	Chooses analysis method	Choice of method changes what is measured; no recorder-window partition perturbation
Feng et al. (2024)	Propositional probes of latent world states; activations	Monitors via probes	Shows recorded layer can diverge from behavior; does not perturb the recording window
Stolfo et al. (2023)	Causal-mediation attribution; internal states	Intervenes on internal states	Attributes mechanism; we do <i>not</i> and report only recorder-level reachability
This work	Identity-lock adjacent difference; active-block event window	Matched block-length intervention plus required allocation	Separates a-priori set composition from from-events magnitude; does not claim mechanism

Table 15: Nearest-neighbor positioning against the closest conceptual measurement works. Each prior operates on a different measured dimension; the table states that relation and the checks already run on the recorder-window period-separation dimension. Artifact paths and SHA-16 prefixes are in Table 11; this is not a mechanism attribution.

Closest conceptual work	Their measured dimension (claim)	Same dim. here?	Testable gap this recorder audit runs (already run)
Jiang et al. (2020)	Prompt-level QA recall/accuracy on factual knowledge; measurement as a lower bound on stored facts	No — conceptual source of our two-layer split; their scale is knowledge retrieval	Differential separator, non-nested canvas-96 probes, and five-arm rank-transform invariance (Table 11)
Sahoo et al. (2026)	Hidden-state probe AUROC/accuracy; confound driven by task format, collapsing after deconfounding	No — deconfounds task format, not the recorder partition	Matched composite partition intervention plus a non-nested canvas-96 instrumentation self-check (Table 11)

Continued on next page

Table 15 continued

Closest conceptual work	Their measured dimension (claim)	Same dim. here?	Testable gap this recorder audit runs (already run)
Feng et al. (2024)	Propositional-probe AUROC for latent world states; recorded layer vs behavior divergence	No — compares layer to behavior, does not perturb the recording window	Fixed model/NFE recorder-window perturbation, paired bootstrap, and rank-sum separation (Table 11)

B REPRODUCIBILITY AND ARTIFACT MAP

Execution cost, shard completion, and event-row validation are reported in Table 16, separately from provenance, so that spend is never conflated with a completeness check. Every quantitative statement in the main text traces to the on-disk artifacts in Table 11; hashes are whole-file SHA-256 prefixes computed from the current workspace.

Determinism. Decoding uses temperature 0 with *remasking=low_confidence*; under the fixed decoding rule and fixed NFE schedule, token selection is deterministic on a fixed canvas, so the identity-lock profile $m(r)$ and its adjacent difference $d(r)$ are deterministic functions of (model, checkpoint, seed, question set, NFE schedule). Min-max separation $\text{sep}(P)$, period selection, and every reported spectral value are deterministic properties of these recorded profiles (only the bootstrap / permutation-replicate quantities carry sampling randomness). Determinism is scoped per (checkpoint, seed, question assignment, NFE schedule); cross-seed variance is not the pre-registered estimand.

Environment. The registered analyses ran under Python 3.11.13, NumPy 2.4.6, pandas 2.3.1, PyArrow 21.0.0, PyTorch 2.7.1; the model is a locally registered LLaDA-8B-Instruct-scale masked dLLM, blockwise decoded over GSM8K (Cobbe et al. 2021), a fixed 300-question canvas at 256 positions and 48 total NFE per arm. The checkpoint is a private local artifact (registry per the project evaluation ledger) and is *not* byte-identical to the public LLaDA-8B-Instruct release; we verify its magnitude is at the same scale but do not assert release equivalence. The full manifest and core-script index are in Table 11.

Cross-environment reproduction. The estimator source was touched after the submission manifest recorded its hash, so rather than assert equivalence we re-derived every arm from raw events with the *current* bytes in a second software environment (Python 3.12.3, NumPy 2.5.2, pandas 3.0.5). All three arms reproduce their sealed sep vectors and selected periods *exactly*, so the second-environment Table 3 values equal the first-environment values bit-for-bit and are not restated as a duplicate column. The complete arm-by-candidate values and both estimator hashes are indexed in Table 11.

Repro kit. Seeds, question identities, the event schema, and a one-command rebuild are provided for the off-GPU parts. Decode uses *seed=20260820* and the frozen question subset $\text{decode_id} \in [345, 645)$. Every numeric readout can be regenerated from the recorded-event schema with the indexed estimator and one-shot re-derivation (Table 11). The model checkpoint cannot be re-released without authorization; an anonymous supplement of releasable events, scripts, and hashes remains subject to provenance approval.

Execution accounting	Recorded value
Hardware	eight GPUs
Timing	330 running s = 0.73 GPU-h; 130 queueing s reported separately and excluded
Shard completion	$L = 16$: 4/4; $L = 64$: 4/4
Event-row check	$L = 16$: 153,600 = 300×512 ; $L = 64$: 499,200 = 300×1664 ; $L = 32$ control: 268,800 = 300×896 rows for <code>decode_id</code> $\in [345, 645)$ of the frozen NFE-48 pool

Table 16: Execution cost and completion checks for the two new treatment arms and the frozen control subset. Cost is reported for the GPU submission that produced the $L=16$ and $L=64$ arms; shard and event-row checks establish completion, whereas the control arm reuses a pre-existing pool.

AI-assisted preparation disclosure. Image generation produced qualitative Figure 1; the selected candidate passed an element-level semantic audit and carries no measurements. Figure 2 remains data-derived. An AI assistant edited prose and LaTeX; all claims, numbers, citations, and hashes were checked against the listed evidence.

C NOTATION AND TERMINOLOGY

Table 17 collects the estimator notation and verdict vocabulary; Table 18 collects the bed names, the treatment variable, and the content-layer terms. Each entry names the section that defines it.

Table 17: Estimator notation and legacy decision-table vocabulary; terms point to their operative definition and do not attribute mechanism.

Term	Meaning (defining location)
$\ell_{i,r}$	Identity-lock time: earliest recorded step after which decode i ’s token at relative position r never changes again (Eq. (1)).
$m(r), d(r)$	Per-position median lock time over decodes, and its adjacent difference $m(r+1) - m(r)$ (Section 2).
$\mathcal{B}_P, \mathcal{I}_P$	Candidate- P boundary set $\{r : r \bmod P = P - 1\}$ and its complement, the candidate interior (Section 2).
$\text{sep}(P), P^*$	Min-max period separation $\min_{\mathcal{B}_P} d - \max_{\mathcal{I}_P} d$, and its maximizing candidate (Eq. (2)).
DETECTED	$\text{sep}(P^*) > 0$, bootstrap modal period = P^* , mode fraction ≥ 0.95 (registered rule, Section 2).
REFUTED / NOT-SUPPORTED	Measured $\text{sep}(P) < 0$ for a candidate that the decision table treated as refutable / for the originally registered intrinsic-32 conjunct, whose branch was withdrawn post-data (Table 6).
A-PRIORI layer	What follows from the estimator’s algebra alone, before data — e.g. boundary-set nesting (Section 2).
FROM-EVENTS layer	What follows only once measured magnitudes exist — e.g. the sign of $\text{sep}(32)$ (Section 2).
Strictly L -supported	Recorded profile with non-degenerate jumps on all L -block boundaries such that no L -multiple’s exclusive-boundary residual outscores L ’s boundary minimum (Section 3.1).
$G(P)$, perm. q_{95} , fam. q_{95}	Normalized DFT magnitude of $d(r)$ at frequency $1/P$; its per-candidate permutation 95th percentile; the family-wise version over all candidates (Section 3.3).

Term	Meaning (defining location)
L16 arm, L64 arm	The two new matched treatment arms at block length 16 and 64, NFE 48 (Table 1).
L32/NFE48 control	Frozen matched positive control at block length 32, NFE 48: the registered <code>decode_id</code> $\in [345, 645)$ subset, $n=300$ (Sections 2 and 3.1).
L32/NFE32 content bed	Separate, differently budgeted bed used only for the content-association diagnostic (Section 3.4).
Partition-plus-required allocation	The composite treatment: changing the block partition necessarily changes the per-block step allocation at fixed total NFE (Table 1).
Measurement-class label	PERIOD-FOLLOWS-BLOCK-LENGTH: a post-data label stating that the detected boundary period follows block length on the tested bed under the active-block recording window; not a mechanism attribution (Sections 1 and 3.1).
BIN8	Restricted null that permutes residualized confidence within 8 equally populated offset-rank bins, preserving offset structure to 4-position granularity (Sections 2 and 3.4).
<code>conf0</code>	Position- r initial confidence at block-commit: the model’s confidence for r at the first recorded step of its block (Section 2).
Block-unit; resolved position	One decode-block of the content bed; a position whose lock time is defined and that never flips after lock. Terminal-resolved units, which present no within-block pairwise order, are excluded (Table 8).
τ, τ_{res}	Median over units of the per-unit rank association between lock time and confidence; the offset-residualized (content-layer) version (Eq. (3)).

Table 18: Bed, treatment, and content-layer terms; the matched setting and separate content bed remain distinct.